Project Proposal

Title: Era-Adjusted Comparative Analysis of ODI Batting Performance (Cricket): Sachin Tendulkar vs. Virat Kohli

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1. Summary of the Project and Goals

This project proposes a statistical comparison of **Sachin Tendulkar** and **Virat Kohli** — two of cricket's greatest ODI batsmen — through a **data-driven, era-adjusted framework**.
Traditional comparisons using raw averages and strike rates are misleading because the game evolved dramatically between Tendulkar's era (1989–2012) and Kohli's era (2008–present).

The primary goal is to determine **which player dominated their respective era more**, using **normalized performance metrics** that account for the changing scoring environment and competition level.

Specific objectives:

- Quantify and compare ODI batting performance across eras using both **absolute** and **relative** metrics.
- Analyze:
    1. **Average runs scored in ODIs across all players** in Sachin's vs. Kohli's eras.
    2. **Average performance of the top 20 batsmen** from each era to benchmark elite standards.
- Develop and visualize **era-adjusted batting averages (EAA)** and **strike rates (EASR)**.
- Present statistical evidence on relative dominance and consistency across time.

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2. Data Description and Sources

The analysis will utilize multiple **publicly available large-scale cricket datasets**, each with detailed ODI-level and player-level data.

Primary Data Sources

1. **ESPN Cricinfo StatsGuru (Web Scrape / CSV Extraction)**
    - Provides match-level and player-level ODI statistics from 1971–present.
    - *URL:* https://stats.espncricinfo.com/ci/engine/stats/index.html
2. **Kaggle Cricket Datasets**
    - *"ODI Player Innings Dataset (1971–2023)"* — ball-by-ball or innings-level data for all players.

- *"Men's ODI Batting Summary Dataset"* — consolidated per-player summaries.
- *URL:* ODI Matches Historical Data (1971–Present) – Kaggle Dataset

3. **Cricsheet Database (Open Source Ball-by-Ball Data)**
   - Contains structured YAML/CSV data of international matches for statistical modeling.
   - *URL:* https://cricsheet.org/
4. **HowSTAT Historical Cricket Database**
   - Archives of player averages, opposition breakdowns, and yearly summaries.
   - *URL:* https://www.howstat.com/cricket/Statistics/Players/PlayerMenu.asp
5. **GitHub Community Repositories**
   - Example: cricket-data and cricpy repositories containing cleaned ODI datasets (1971–2024).
   - *URL:* https://github.com

Together, these datasets will provide enough granularity to compute yearly averages, top-20 benchmarks, and adjusted metrics.

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3. Preliminary Data and Context

Era Definition

- **Sachin Tendulkar Era:** 1989–2012
- **Virat Kohli Era:** 2008–Present

Approximate Aggregate Comparison (as of 2024)

Player	Matches	Innings	Runs	Average	Strike Rate
Sachin Tendulkar	463	452	18,426	44.83	86.23
Virat Kohli	292+	280+	13,500+	57.3	93.5

Era Context Metrics (Indicative Averages)

Metric	Sachin Era (1989–2012)	Kohli Era (2008–Present)
Avg ODI Team Score	235–260	270–310

Metric	Sachin Era (1989–2012)	Kohli Era (2008–Present)
Global Batting Avg (All Players)	~27.5	~32.1
Avg of Top 20 Batsmen	~38.5	~44.7
Avg Run Rate (per over)	4.6	5.4

These contextual metrics highlight that modern ODIs have become more batting-friendly — justifying the need for **era normalization**.

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4. Methodology and Comparative Framework

The comparison will follow a structured analytical framework:

1. **Raw Performance Metrics**
    - Total runs, 100s, 50s, averages, strike rates.
    - Performance by opposition, venue, and batting position.
2. **Era-Adjusted Metrics**
    - *Era Adjusted Average (EAA)* = Player's Average ÷ Global Average (same year).
    - *Era Adjusted Strike Rate (EASR)* = Player's SR ÷ Global SR (same year).
    - *Top-20 Normalized Score (T20N)* = Player's Average ÷ Average of Top 20 Batsmen (same era).
3. **Time-Series Analysis**
    - Rolling 3-year and 5-year averages for consistency tracking.
    - Trend lines of EAA and EASR across career phases.
4. **Visualization Tools**
    - Line plots for performance evolution.
    - Density plots for distribution of runs per innings.
5. **Statistical Testing**
    - Regression modeling of runs scored against year, opposition, venue, and player.

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5. Preliminary Data Exploration

Preliminary distributions are expected as follows:

- **Runs per innings:** Heavily right-skewed for both players (many low scores, few large innings).
- **Strike rate:** Normally distributed around each player's mean, with Kohli's curve slightly higher.
- **Era averages:** Upward trend over time, confirming that Kohli's era is statistically more run-rich.

Early exploration from Kaggle and Cricinfo datasets confirms:

- Missing values <5%, mostly in early-1990s matches (balls faced, venues).
- Consistent units across datasets: runs (integer), strike rate (numeric), dates (YYYY-MM-DD).
- Averages to be considered only for players with more than 20 matches played.

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6. Challenges and Limitations

- **Incomplete Historical Data:** Detailed ball-by-ball data for 1980s–1990s ODIs may be partially missing.
- **Contextual Factors:** Pitch and match conditions not fully captured numerically.
- **Cross-Era Bias:** Evolution of powerplays, T20 influence, and modern fitness affect scoring rates.
- **Normalization Accuracy:** Yearly averages may fluctuate due to small-sample anomalies in early data.

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7. Areas of Inquiry and Techniques

Research Questions

1. Who demonstrated greater dominance relative to their era's average and elite peers?
2. How did era-based changes in run rates and strike rates affect perceived performance?

Planned Analytical Techniques

- Descriptive and inferential statistics.
- Ratio-based normalization.
- Visualization using ggplot2 (R).
- Comparative trendlines for "Top 20 Batsmen Average vs. Player Average."

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8. Expected Outcomes

- A quantitative, era-adjusted comparison of Tendulkar and Kohli's ODI batting impact.
- Empirical insights into how run-scoring inflation and top-tier benchmarks influence perception.
- A nuanced conclusion answering *"Who was more dominant in their time?"* supported by data.

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9. References

1. ESPN Cricinfo StatsGuru. (2024). *Cricket Statistics and Records Database*. https://stats.espncricinfo.com/
2. Kaggle Datasets. (2023). *ODI Player Innings and Batting Summary Datasets*. https://www.kaggle.com/
3. Cricsheet. (2024). *Open Source Cricket Data Repository*. https://cricsheet.org/
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5. GitHub – cricket-data repositories. (2023). *Open Source Cricket Analytics Data*.

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